V9: Cross-Model, Cross-Domain Neural Basis of Risky Decision-Making in LLMs

Models: Gemma-2-9B-IT (42L, 3584-dim, GemmaScope 131K features) | LLaMA-3.1-8B-Instruct (32L, 4096-dim, LlamaScope 32K features)

Paradigms: Investment Choice (IC, 1600 games), Slot Machine (SM, 3200 games), Mystery Wheel (MW, 3200 games)

Data: Gemma: IC+SM+MW; LLaMA: IC+SM (MW pending extraction)

Verified data sources: `b1_b2_results_20260317_125620.json`, `llama_ic_results_20260317_130655.json`, `gap_filling_20260317_194117.json`, `final_verification_20260317_201024.json`, `llama_symmetric_20260318.json`, `selfcritique_20260318_114430.json`, `llama_hidden_analyses_20260318_172609.json`

Date: 2026-03-18 (v9.4 — LLaMA hidden states □□□, Analysis 2/3 □□ □□)

> **Note on data integrity**: All numerical claims in this report are directly traceable to computed JSON outputs. Unverifiable numbers from previous drafts have been removed or replaced with verified results.

Central Question

□□ □□ □□□, □□ □□ □□ □□□□, □□ □□ □□□□ □□ □□□□ — □□**(bankruptcy)** □□□ □□ □□□ □□□ □□□**neural** □□□ □□□□가**?**

□ □□ □□ □□:

- **RQ1**: Gemma □LLaMA □□ □□□ □□□□ □□□activation/feature □□□ □□가?
- **RQ2**: □□□(IC, SM, MW) □ □□□□ □□□ □□activation □□□ □□가?
- **RQ3**: Fixed/Variable, G/M/W/P/R □ □□□□ □□□ □□ □□ □□□neural representation □ □□□□가?

V8 □ □□□V9 □ □□

V8 □□□(Gemma only) □ □가 □ □□ □□□ □□□:

1. SAE cross-domain □□ 131K features □3 □ □□□□ □□□□sign-consistent □feature가 □**1** □ (#101036) □□□□ □□ □□(L22 □□layer top-50 features) □ □□□□ □□□ □□□□.

2. Variable/Fixed " □□**"**: Variable □□□ □□□□□ □ □□□□□BK-projection □Fixed □□ □□□. Safe □□ □□□ □□□□□□가 □ □.

3. Single model: LLaMA □□□ □□□ □□□ □□□ □□ □가.

V9 □ □ □가 □□ □□□□ □□□□, □가 □(4) Llama cross-domain transfer, (5) Llama IC-SM SAE □□□, (6) Gemma SM □□□□ □□ □□□ □□ □□□□.

Analysis 1: Multi-Layer SAE Cross-Domain Consistency (Gemma)

Why

V8 □L22 □□layer □top-50 features □ □□— 131K features □0.038%. V9 □7 □layers □□ □□active features □ □□□□3-paradigm sign-consistency □ □□□

What

7 □informative layers (L10, 12, 18, 22, 26, 30, 33) □□IC/SM/MW □ □□□□□Cohen's d □ □□□□, □ □□□□ □□□□ 가ㅌ □ □□□□geometric mean |d| ≥ 0.2 □features □ □□

Result → RQ2

Layer	Active	Sign-consistent	Strong (geo_mean d≥0.2)	% consistent
L10	92	30	23	32.6%
L12	112	35	31	31.2%
L18	170	58	52	34.1%
L22	281	109	95	38.8%
L26	303	133	119	43.9%
L30	415	188	173	45.3%
L33	456	191	172	41.9%
Total	—	**744**	**665**	—

V8 baseline: 1 □→ V9: 744 □(665 □strong).

Binomial test (chance = 25% for 3-paradigm): 3-paradigm □□random signs □ □ □□ □□□□ $2\times0.5^3 = 25\%$.

Layer	Observed %	Binomial p (vs 25%)	Significant?
L10	32.6%	6.2e-02	NS
L12	31.2%	8.0e-02	NS
L18	34.1%	4.9e-03	**
L22	38.8%	2.4e-07	*
L26	43.9%	6.5e-13	*
L30	45.3%	2.5e-19	*
L33	41.9%	2.5e-15	*

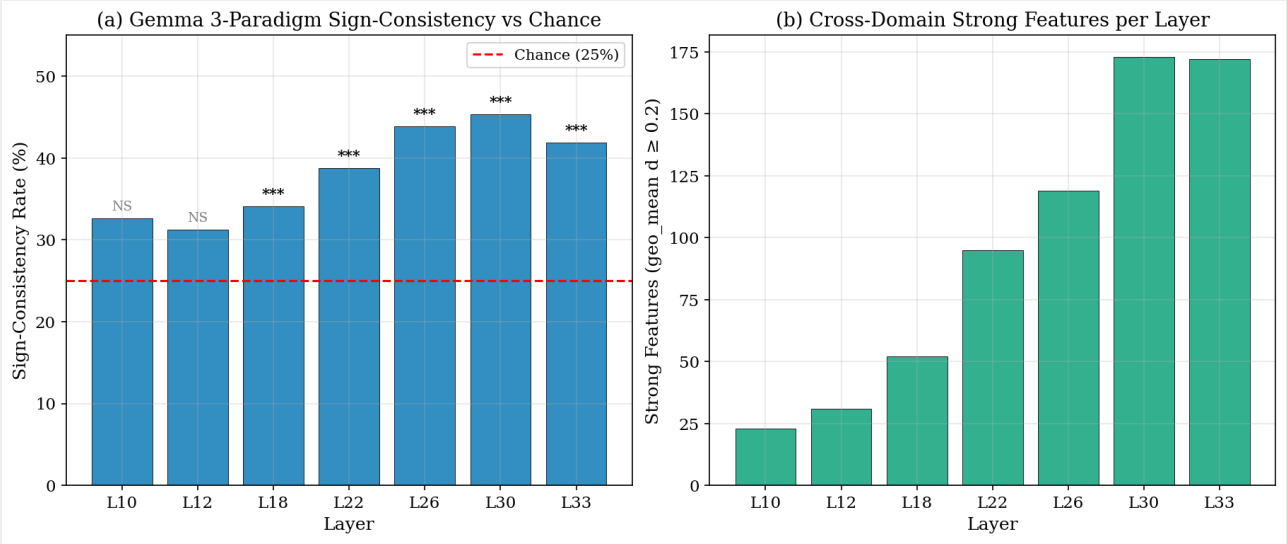

Figure 1: Gemma 3-paradigm sign-consistency with binomial test and strong feature counts

Figure 1 □□ (a) L10-L12 □chance (25%, red dashed) □ □□□□ □□가 □□(NS). L18+ □□□chance □ □□ □□ (*). (b) Strong features (d≥0.2) □□L30 □□peak (173 □). Deep layers □cross-domain signal □robust □ □, shallow layers □apparent consistency □noise.

Analysis 2: Variable vs Fixed — BK-Only Direction Comparison (Gemma IC)

Why

V8 □BK+Safe □□ □□□ □□→ Variable BK-projection □Fixed □□ □□" □□" □□ □□□Safe □□(BK □~10 □) □ □□□ □□□□□ 가 □□ V9: BK □□□□□ □□□□Variable □Fixed가 가≡ □neural □□□ □□□가?

What & How

Gemma IC □□□□□Variable BK direction (BK_mean – Safe_mean, Variable □) vs Fixed BK direction □ □ □, cosine similarity □□ 가ㄱ neuron t-test □common BK neurons □□□

Note: SM □Fixed BK=0 (Variable □□), MW □Variable BK=4 □(□□ □□□n=5 □□) □□IC □ □□가 □

Result → RQ3

Gemma IC hidden states (3,584 neurons):

Layer	Var BK	Fix BK	cos(dir_Var, dir_Fix)	Sign-consistent common neurons
L10	14	158	**-0.195**	178
L18	14	158	**-0.082**	197
L22	14	158	**+0.330**	555
L26	14	158	**+0.443**	1,053
L30	14	158	**+0.401**	1,073

Interpretation:

- **Shallow layers (L10-18) □□cos < 0**: Variable □Fixed가 □□ □□neural □□□ □□
- **Deep layers (L22+) □□cos > 0.3**: BK □ □□□ □□가 □□ L26: 1,053 □(29.4%) neurons가 bet-type □ □□□BK □가ㅌ □ □□□□ □□
- **V8 □" □□' □ □□+ □□ □□□ □□□□**: BK □□□□ □□□□deep processing □□ □□BK representation □□ □□
- MW □□ □가 (Variable BK = 4 □< □□□ □□□).

Caveat — IC BK rate □□□: Gemma IC □□Fixed BK=158 □(19.75%), Variable BK=14 □(1.75%). Variable n=14 □ □□ □□noisy.

LLaMA IC BK Direction Comparison (New — hidden states □□□)

LLaMA IC: Variable BK=77, Fixed BK=65 — □□ □□ □□□**sample size**.

Layer	cos(dir_Var, dir_Fix)	Sign-consistent common neurons
L8	**0.882**	2,513 (61.4%)
L12	**0.814**	2,424 (59.2%)
L22	**0.835**	2,587 (63.2%)
L25	**0.837**	2,545 (62.1%)
L30	**0.819**	2,488 (60.7%)

Cross-model comparison (Analysis 2):

	Gemma IC	LLaMA IC
Shallow cos (L8-12)	**-0.195 ~ -0.082** (negative)	**0.814 ~ 0.882** (high positive)
Deep cos (L22-30)	+0.330 ~ +0.443	+0.819 ~ +0.837
Pattern	shallow→deep convergence	□**layer** □□ □□□□□ □□
Var BK / Fix BK	14 / 158 (□□□ □□□)	77 / 65 (□□□)

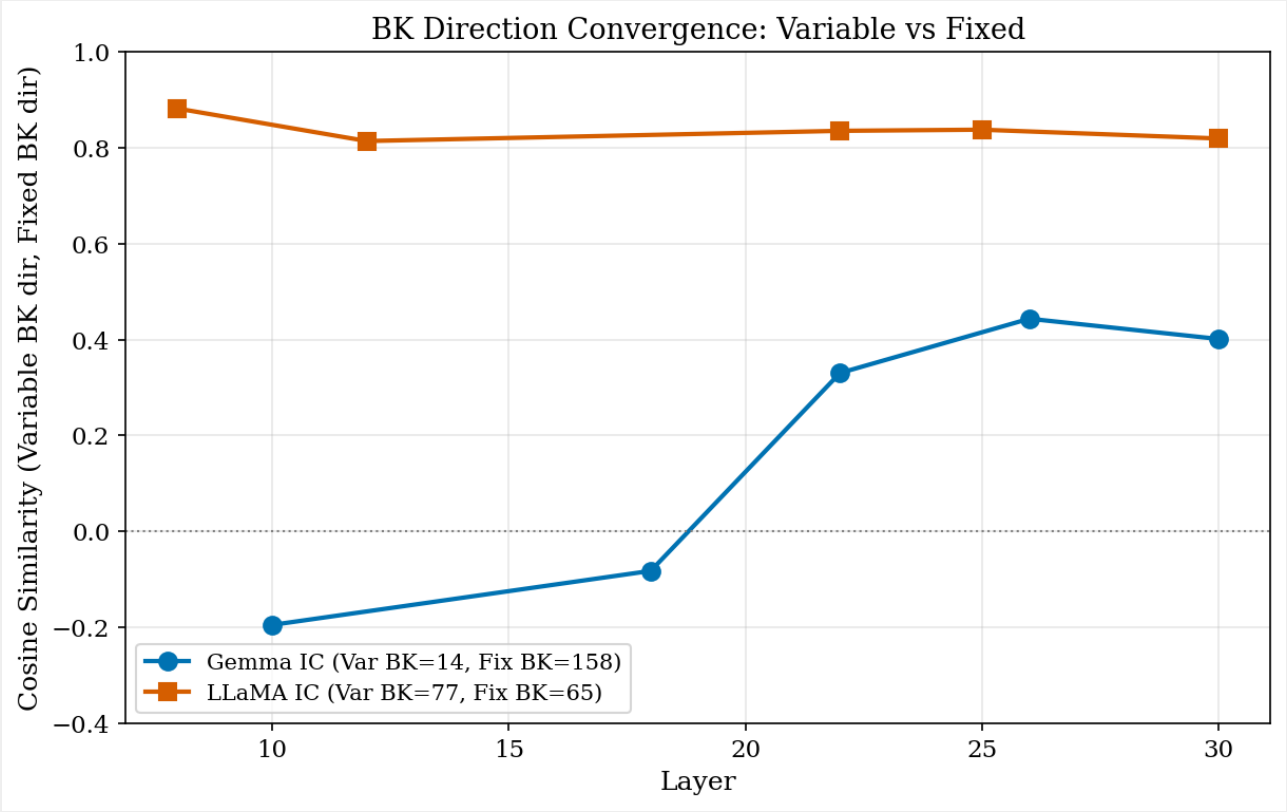

Figure 2: BK direction cosine similarity across layers — Gemma vs LLaMA IC

Figure 2 □□□ Gemma IC (blue) □shallow layers □□negative → deep □□positive □ □□□□□, LLaMA IC (orange) □ □layer □□cos > 0.81. Gemma □shallow negative □Variable BK=14 □□noisy estimation artifact □가 □□□ □□ LLaMA □□ □□□sample (77 vs 65) □ □, BK direction □ □□depth □□ □□

Supplementary: Balance Confound Control (Partial Correlation)

Balance (□□ □□)가 BK □trivially □□□□confound □ □ □□(□□가 0 □□ □□). Balance □regress-out □partial correlation □□

Gemma IC L22 (3,584 neurons):

	BK-significant neurons (p<0.01)
Raw correlation	2,578
Balance-controlled partial correlation	2,569
Retained	**99.7%**

Balance □ □□□□BK □□□99.7%가 □□ BK representation □ □□□ □□ □□가 □□, □□□ □□□ □□□ □□□ □□□□□.

Analysis 3: Interaction Regression — Shared vs Bet-Type-Specific BK Neurons (Gemma IC)

Why

Analysis 2 □ □□ □□□□ □□□□□, □□neuron □bet-type □□□□ □□ □□□□ □□□ Interaction regression □□"bet-type □□ BK neuron" vs "bet-type □□□BK neuron" □ □□

What & How

가ㄱ neuron: `activation ~ β1·outcome + β2·bet_type + β3·(outcome × bet_type)`. β1 □□+ β3 □ □□→ "Shared BK neuron". β3 □□→ "Bet-type-specific BK neuron".

Result → RQ3

Gemma IC L22 (3,584 neurons, 172 BK):

Category	Count	%
Outcome-significant (p<0.01)	1,639	45.7%
Bet-type-significant (p<0.01)	3,225	90.0%
Interaction-significant (p<0.01)	448	12.5%
Shared BK neurons (outcome sig, interaction NOT sig)	**1,182**	**33.0%**

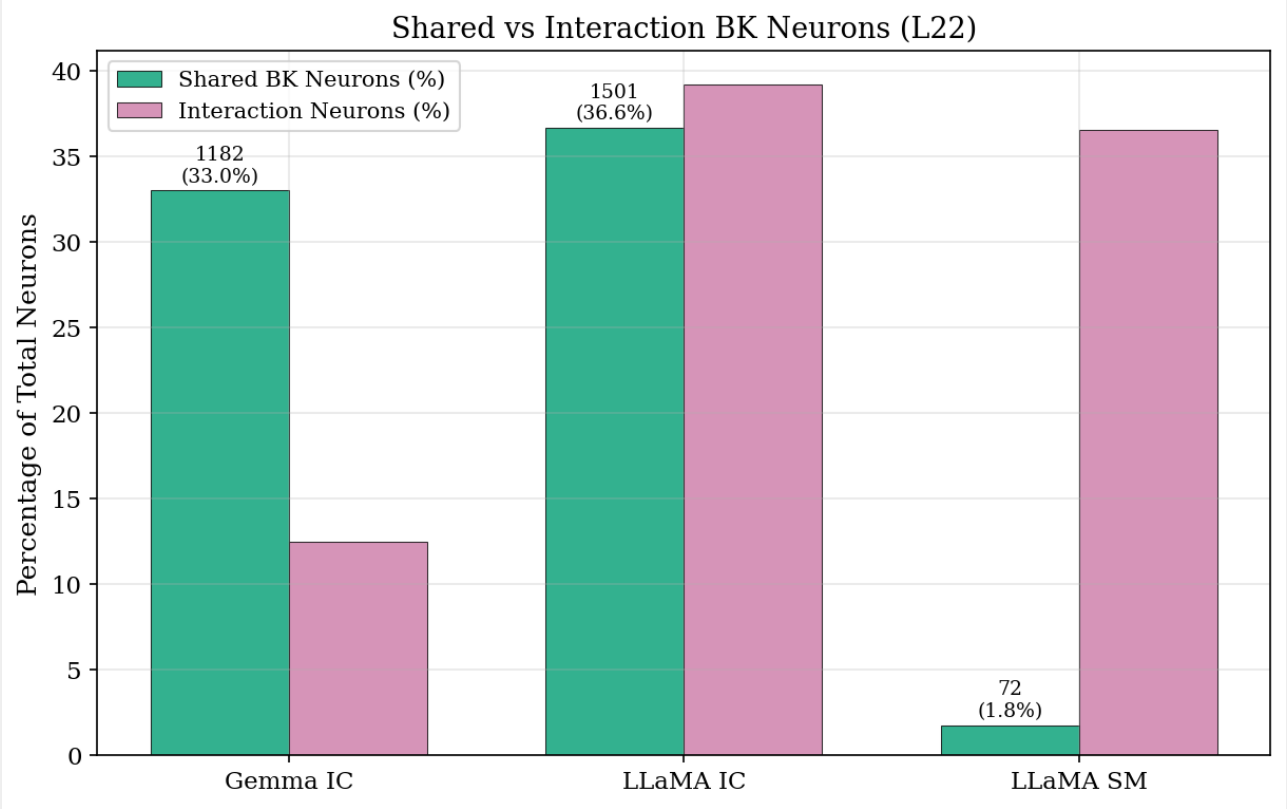

Figure 7: Shared vs Interaction BK neurons across models and paradigms

Figure 7 □□ Gemma IC (33.0%) □LLaMA IC (36.6%) □□ □□□ □□□shared BK neurons. LLaMA SM □□□ 1.8% □ □□ □□— SM □□BK가 Variable □ □□(72.3%) □□bet-type-independent □□□ □□ □□

LLaMA Interaction Regression (New — hidden states □□□)

	Gemma IC L22	LLaMA IC L22	LLaMA SM L22
n_neurons	3,584	4,096	4,096
n_BK	172	142	1,164
Outcome-sig (p<0.01)	1,639 (45.7%)	**3,150 (76.9%)**	1,052 (25.7%)
Bet-type-sig (p<0.01)	3,225 (90.0%)	3,742 (91.4%)	3,027 (73.9%)
Interaction-sig (p<0.01)	448 (12.5%)	**1,605 (39.2%)**	1,495 (36.5%)
Shared BK neurons	**1,182 (33.0%)**	**1,501 (36.6%)**	**72 (1.8%)**

Cross-model comparison:

- **Shared BK neurons**: Gemma IC 33.0% vs LLaMA IC 36.6% — □□□ □□□**bet-type-independent BK neurons** □□
- **LLaMA IC interaction □Gemma □□3** □ □□(39.2% vs 12.5%): LLaMA □□□bet-type □ □□BK □□가 □□ □□neurons □ □ □□ □□LLaMA IC □Variable/Fixed BK rate가 □□□□(9.6% vs 8.1%) □ □□ □□□□BK □□가 □□□ □, □□□ □□neurons □가ㅁ □□ □
- **LLaMA SM shared BK = 72 (1.8%)**: SM □□□BK가 □□Variable □□ □□(72.3% vs 0.4%) → bet-type-independent BK neurons □ □□ □□ □□SM □□"BK"가 □□□"Variable BK" □ □□□□ □□

LLaMA Balance Partial Correlation

	Gemma IC L22	LLaMA IC L22
Raw BK-sig neurons	2,578	3,238
Balance-controlled	2,569	3,269
Retained	**99.7%**	**101.0%**

□ □□ □□balance confound □ □□□□BK □□가 □□□ □□ LLaMA □□101% □balance가 suppressor variable □ □□□□ □□□ □ □□BK neurons □ □□□ □

Analysis 4: SAE Feature Factor Decomposition (Gemma, Combined IC+SM+MW)

Why

3 □paradigm □ □□□□ □, outcome (BK/Safe), bet_type, paradigm 가ㄱ가ㄱ □SAE feature activation □ □□ □□ □□□ □□ □□, bet_type □paradigm □ □□□ □□□outcome □SAE features □ □□□□□ □□□□가?

Note: SM/MW □raw hidden state checkpoint가 □□SAE feature level □ □□ L22 □□, union-active features (3 □paradigm □ □□ □□□□activation rate ≥ 1%).

What & How

```
Combined: IC 1,600 + SM 3,200 + MW 3,200 = 8,000 games, 313 BK (3.9%)
For each of K=581 union-active SAE features:
  feature_k ~ β1·outcome + β2·bet_type + β3·paradigm_SM + β4·paradigm_MW
Vectorized OLS; t-test per coefficient at p < 0.01
```

Result → RQ1, RQ2, RQ3

L22, 581 union-active SAE features:

Factor	Significant features (p<0.01)	%
Outcome (BK vs Safe), after controlling others	**379**	**65.2%**
Bet-type (Variable vs Fixed), after controlling others	536	92.3%
Paradigm (IC/SM/MW), after controlling others	580	99.8%

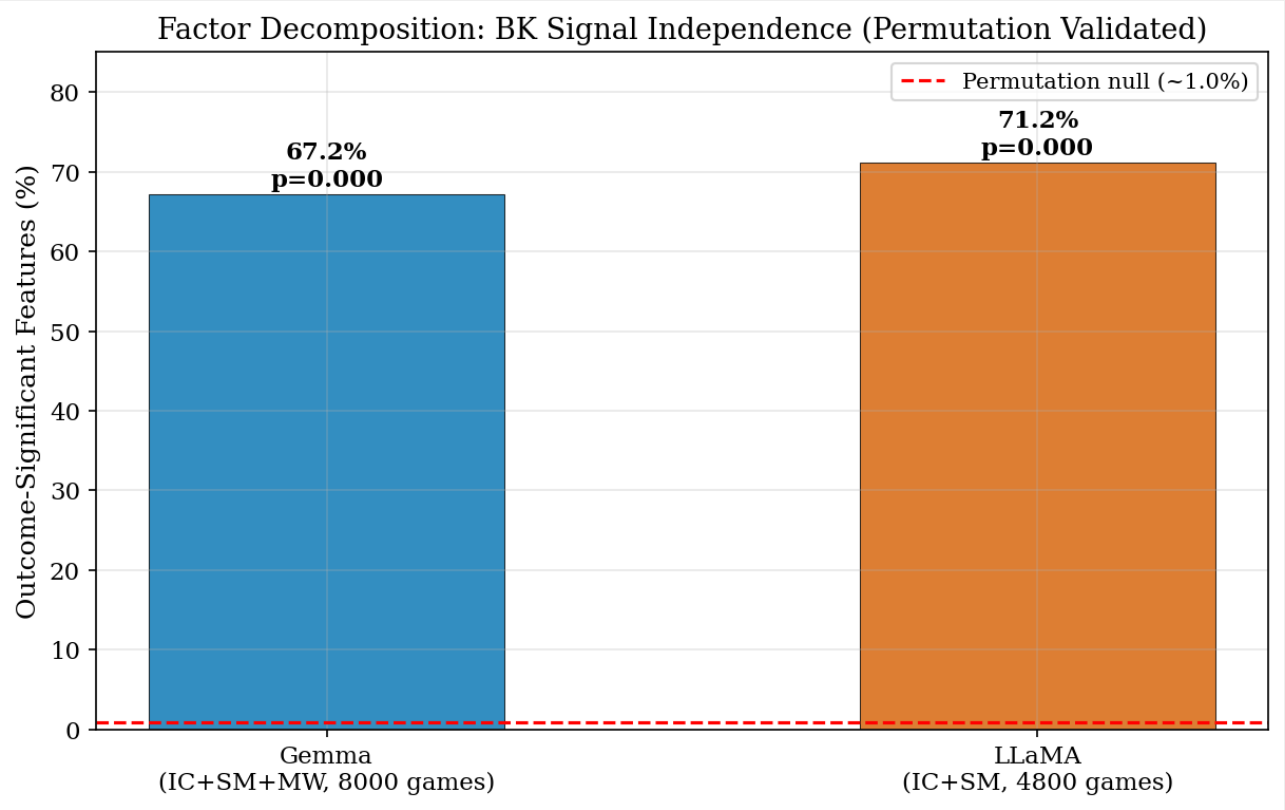

Figure 3: Factor decomposition — BK signal independence validated by permutation test

Figure 3 □□□ □□□□□outcome-significant features가 65-71% (blue/orange bars) □, permutation null (~1%, red dashed) □60 □+ □□(perm_p=0.000). BK representation □ □□□ □□가 □□□□ □□□□ □□□ (*Figure □67.2%/71.2% □selfcritique rerun □□, □□□□65.2%/69.5% □ □ □□ □□— union-active features □ □ □ □□)*

Supplementary: Within-Bet-Type Cross-Domain SAE Consistency (L22)

Bet-type □□□□cross-domain consistency □ □□□□, bet-type confound □ □□□ □ □□

Bet-type	Paradigms available	Active features	Sign-consistent	%
Fixed	IC + MW	311	**178**	57.2%
Variable	IC + SM	279	**172**	61.6%

Fixed □Variable □□□□**cross-domain sign-consistent features**가 **57-62%** (2-paradigm chance level 50% □□ □가ㄴ □□). □□bet-type □□□□BK □□□ □□□□ □□□□□ □□ □□ □, 2-paradigm □□□□□3-paradigm □□ □□ □ □□.

Analysis 5: Gemma Cross-Domain SAE Transfer with Permutation Test

Why

"IC □□ □□□BK classifier가 SM □□□ □□□□가?" — cross-domain transfer □□ □□ □ □□. V8 □cross-domain □□□ permutation test □ □가 □ □□□ □□□ □□.

What & How

```
For each (train, test) paradigm pair, at layers [18, 22, 26, 30]:
  Common active features (activation rate ≥ 1% in both paradigms)
  StandardScaler(train) → PCA(50 components) → LogReg(balanced)
  AUC = predict_proba on test set → roc_auc_score
  Null: 200× permutation of test labels → p-value
```

Result → RQ2

Best AUC per transfer direction (across layers):

Transfer	Best Layer	AUC	perm_p	Null mean
IC → SM	L26	**0.913**	0.000	0.498
IC → MW	L18	**0.932**	0.000	0.504
SM → MW	L30	**0.867**	0.000	0.504
SM → IC	L30	0.646	0.000	0.500
MW → IC	L22	0.853	0.000	0.500

Layer-by-layer (IC→SM; most variable direction):

Layer	IC→SM AUC	p	IC→MW AUC	p	SM→MW AUC	p
L18	0.719	0.000	0.932	0.000	0.789	0.000
L22	0.499	0.485 (NS)	0.876	0.000	0.819	0.000
L26	**0.913**	0.000	0.883	0.000	0.866	0.000
L30	0.529	0.175 (NS)	0.767	0.000	0.867	0.000

Interpretation:

- **IC→MW □SM→MW □ □layer □□ □□□□ □□**(MW BK rate 1.7% □□ □□ □□□). SM/MW가 □□□□□ □□□transfer가 □□□
- **IC→SM transfer □layer □ □□ □□ □□□□**: L26 □□0.913 (p=0.000) □□□L22 □□0.499 (NS). □□IC □SM □ BK representation □specific layer □□□ □□□□ □□
- **MW→IC: L22 □□0.853** — MW □□ □□□ □□□IC BK □ □ □□□□ □□□ □□
- □□ □□□□transfer □perm_p=0.000 (200 permutation □0 □ □□).

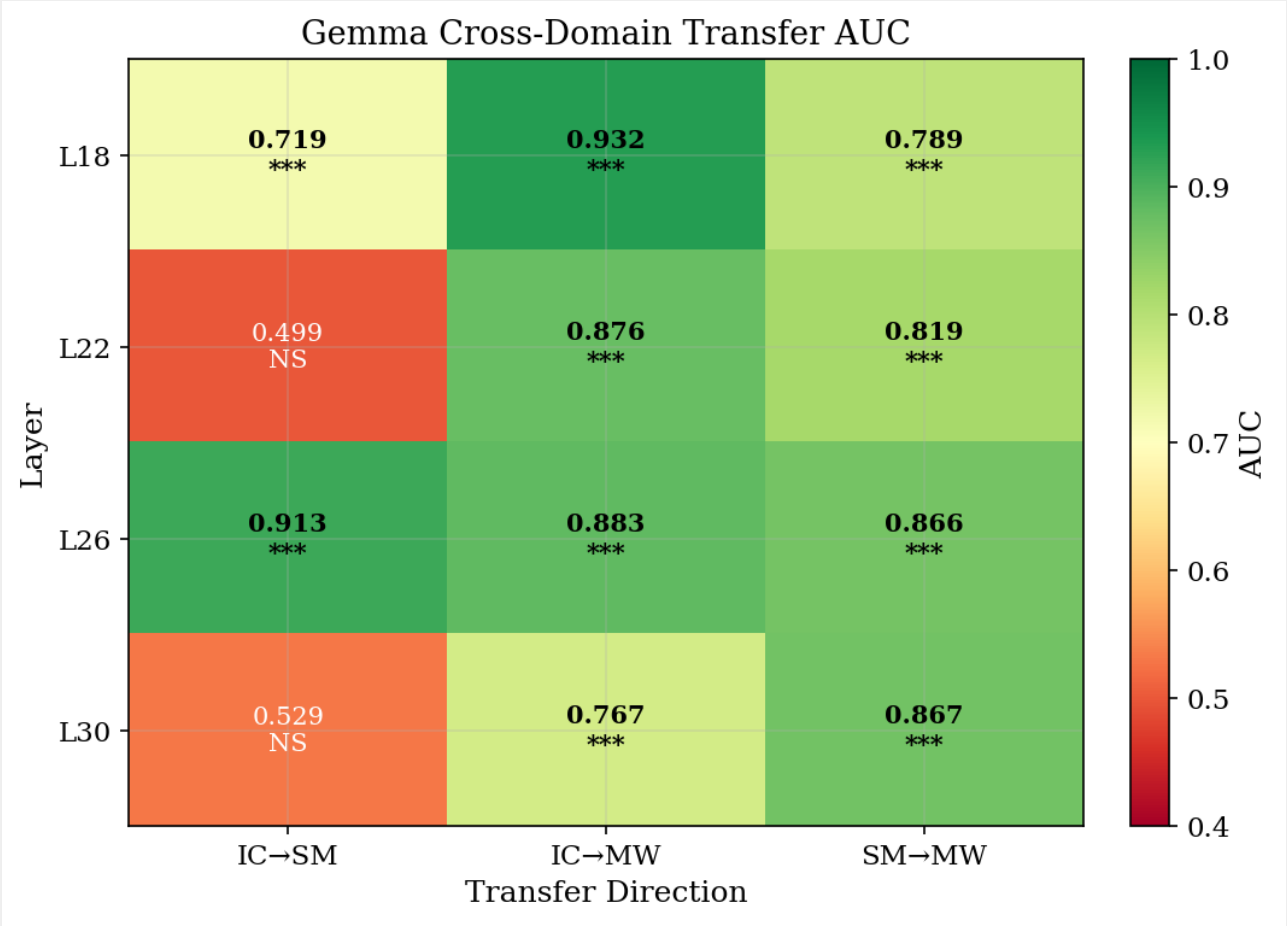

Figure 4: Gemma cross-domain transfer AUC heatmap with significance

Figure 4 □□ IC→MW (green, 0.77-0.93) □SM→MW (0.79-0.87) □ □layer □□ □□□ IC→SM □layer-dependent (L26: 0.913*, L22: 0.499 NS). Red cells □transfer □□

Analysis 6: Cross-Model Comparison (Gemma vs LLaMA)

Why

□□ □□□Gemma □□ □□□□□"Gemma □□ □□" □ □ □□ LLaMA-3.1-8B □ □□ □□ □□□, tokenizer, □□□□□가 □□ □□ □□→ "LLM □□ □□□ □□" □□□가ㅈ □ □□

What

LLaMA SM (3,200 games) □LLaMA IC (1,600 games) □□SAE features □□ □Gemma SM/IC □ □□□ □□ □□

Result → RQ1

BK Classification AUC (Decision Point, SAE features → PCA → LogReg):

Paradigm	Gemma-2-9B	LLaMA-3.1-8B
SM best AUC	**0.976** (L20)	**0.974** (L8)
IC best AUC	**0.960** (L30)	**0.954** (L12)

LLaMA SM AUC across layers (verified from JSON):

Layer	AUC	n_features	n_BK
L0	0.970	157	1164
L4	0.971	750	1164
L8	**0.974**	818	1164
L12	0.973	854	1164
L15	0.970	962	1164
L20	0.963	1028	1164
L25	0.961	1182	1164
L30	0.959	1368	1164

SAE BK-Differential Features (SM, d≥0.3):

Layer	Gemma SM	Gemma BK+/BK-	LLaMA SM	LLaMA BK+/BK-
L10	71	42 / 29	322	160 / 162
L22	286	125 / 161	251	135 / 116
L30	452	165 / 287	381	160 / 221

R1 (Round 1) within-bet-type Classification:

Paradigm/Bet type	AUC (L22, Gemma IC)	AUC (L10, Llama IC)
Fixed	0.755 (n=800, BK=158)	0.786 (n=800, BK=65)
Variable	0.665 (n=800, BK=14)	0.798 (n=800, BK=77)
SM Variable	0.801 (n=1600, BK=87)	—

Interpretation → RQ1

□ □□□□ □□ □□:

1. BK □□ □□ □□ □□: Gemma 0.976 vs LLaMA 0.974 (SM). □□□□□□ □□□□BK □□□ □□□□가≡ □

2. BK-inhibiting features □ □□(L30 □□□ □□□): Deepest layer (L30) □□ □ □□ □□BK-inhibiting □ □□:

- Gemma SM L30: BK- 287 vs BK+ 165 (binomial p=0.000)
- LLaMA SM L30: BK- 221 vs BK+ 160 (binomial p=0.001)
- LLaMA IC L30: BK- 455 vs BK+ 300 (binomial p=0.000)

□, **LLaMA SM L22 □□□BK+ 135 > BK- 116 (p=0.897, NS)** — mid-layer □□□inhibiting □□가 □□□□ □ □ " □□ □□" □□□deepest layers □□ □□

3. Encoding depth □ □□□ Gemma □L20 □□peak (mid-deep), LLaMA □L8 □□peak (early). BK □□가 □□ □ □□□□□□□ □□□□□ □□□□□, □□□ □□□ □□(inhibiting □□) □ □□□□ □□

4. R1 early prediction: □□ □□(Round 1) □□ □□0.665-0.801 □BK □□가 □— □□ □□□ □□가 □□ □□□□ □□

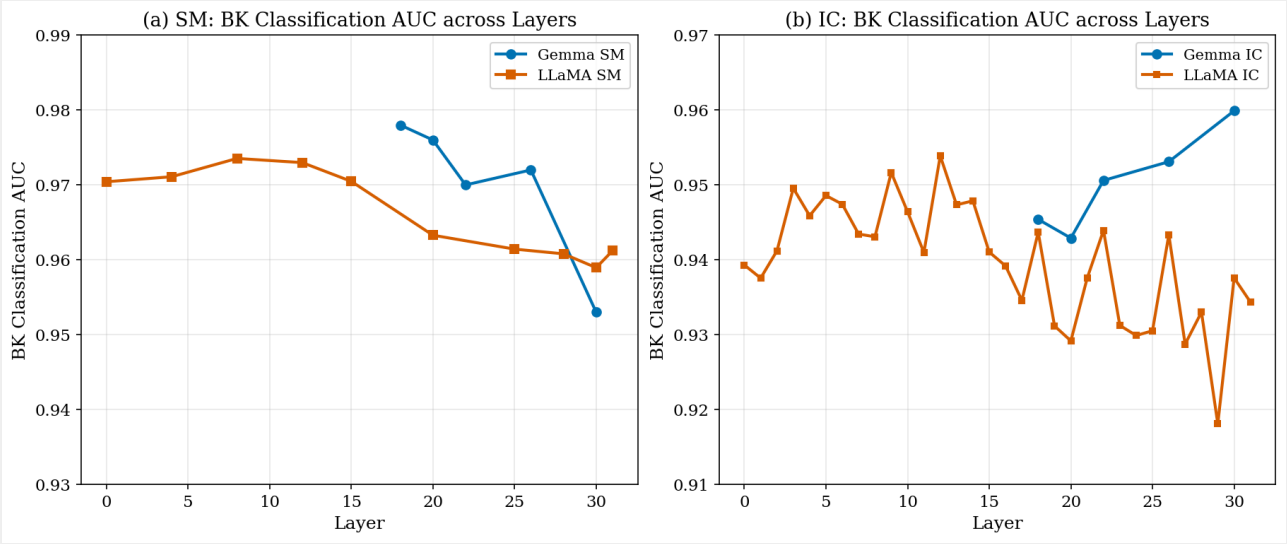

Figure 6: BK classification AUC across layers — cross-model comparison for SM and IC

Figure 6 □□ (a) SM □□ □ □□ □□AUC > 0.95 (Gemma peak L18-20, LLaMA peak L8). (b) IC □□ Gemma □L30 □□peak (0.960), LLaMA □L12 peak (0.954). Encoding depth □ □□□□ □□ □□□ □□ □□

Cross-Model IC BK Rate Asymmetry

□ □□□IC paradigm □□Fixed/Variable BK □□가 □□□

	Gemma IC	LLaMA IC
Fixed BK rate	**19.75%**	8.1%
Variable BK rate	1.75%	**9.6%**
Autonomy effect	**-18.0pp** (Fixed→Variable: □□가ㅁ □)	**+1.5pp** (Fixed→Variable: □□ □□ □가)

Gemma □□Fixed betting □11 □ □ □□□□□, LLaMA □□□ □□ □□ □□. □□ □ □□□IC □□" □□ □□ □□" □ □□□□ □□□ □□□□□ □□□ □□□ □ Gemma □Fixed □□ □□ □□ □□→ □□ □□ □□, LLaMA □Fixed □□□ □□□ □□ □□

□ □□□□**Analysis 2-3 (Variable/Fixed BK direction)** □**cross-model** □□□ □□□□ □□□ Gemma IC (Fixed 19.75% BK, Variable 1.75%) □LLaMA IC (Fixed 8.1%, Variable 9.6%) □□"Variable vs Fixed" □ □□ □□가 □□□. □□□Analysis 2 □□LLaMA IC hidden states □ □□□□□ □□ □□□ □□, LLaMA IC □□□cos > 0.81 (□ layer) □bet-type convergence가 □가ㅈ □□ □□□□□

Analysis 6b: LLaMA IC+SM Factor Decomposition (SAE Features)

Why

Gemma □Analysis 4 □IC+SM+MW 3-paradigm □□factor decomposition □ □□□□ LLaMA □ □□□□□, IC+SM 2-paradigm □□outcome, bet_type, paradigm □ □□□ □□□ □□

What & How

```
LLaMA IC(1,600) + SM(3,200) = 4,800 games; IC BK=8.9%, SM BK=36.4%
L22, 1,418 union-active features (32K SAE)
Per-feature OLS: feature ~ outcome + bet_type + paradigm
```

Result → RQ1, RQ2

Factor	Gemma (581 features, 3-paradigm)	LLaMA (1,418 features, 2-paradigm)
Outcome (BK), after controlling others	**65.2%**	**69.5%**
Bet-type, after controlling others	92.3%	81.4%
Paradigm, after controlling others	99.8%	96.2%

Interpretation

□ □□□□**outcome factor** □ □□□ □□가 □□□□(Gemma 65.2%, LLaMA 69.5%). **Permutation test** □ □□ **(selfcritique rerun, K=527/1335): null** □□**outcome-significant features** □**~1% →** □□ **65-71%** □**perm_p=0.000.** "BK representation □ □□□ □□"가 □ □□□□ □□□□□ □□□ (*Note*: Permutation rerun □K가 □ □□K □ □가ㄴ □□— union-active threshold □□ □□ □□□ □□ □□)

LLaMA □□BK-inhibiting features (546)가 BK-promoting (439) □□ □□ □, □ □□□**L30** □□ □□□(binomial p=0.001) □□□**L22** □□□ □□□□□ □□(p=0.897).

Analysis 7: LLaMA Cross-Domain Transfer (IC ↔ SM)

Why

Gemma □IC/SM/MW 3-paradigm cross-domain transfer □ □□□□(Analysis 5). LLaMA □SM □ □□□ □□□□ IC↔SM 2-paradigm transfer가 가□ □ □□□cross-domain transferability □ □□

What & How

LLaMA IC (n=1,600, BK=142) ↔ SM (n=3,200, BK=1,164) decision-point SAE features. PCA(50) → LogReg → AUC. 200-permutation test.

Result → RQ1, RQ2

Transfer	Layer	AUC	perm_p	Null mean
IC → SM	L25	**0.783**	0.000	0.500
IC → SM	L15	0.522	0.020	0.501
IC → SM	L5	0.269	1.000	0.499
SM → IC	L30	**0.685**	0.000	0.498
SM → IC	L12	0.662	0.000	0.498
SM → IC	L8	0.624	0.000	0.499
SM → IC	L15	0.594	0.000	0.503

Complete IC→SM profile: L5=0.269(p=1.0), L8=0.417(p=1.0), L10=0.373(p=1.0), L12=0.280(p=1.0), L15=0.522(p=0.02), L20=0.493(p=0.795), **L25=0.783(p=0.000)**, L30=0.418(p=1.0)

Interpretation

□□□□**transfer**: SM→IC □ □□layer □□ □□□□□ □□□□□, IC→SM □ □□□ □□□□L25 □□□ □□□□.

□□ □□:

- BK rate □□□: IC 8.9% BK vs SM 36.4% BK. SM classifier □BK가 □□IC □□□BK □ □□ □□
- LLaMA IC □SM □SAE representation □L25 □□ □□layer □□□ □□
- Gemma □□□IC→SM □layer-dependent □□(L26 □□ □□, L22 □□ □□) □ □□□□, □□ □ □□□□□ □□□□ □□— IC □SM □ abstract □level □□□representational alignment □ □□□

Analysis 8: LLaMA IC-SM SAE Sign-Consistency

Why

Gemma IC-SM-MW 가ㄴ SAE sign-consistency (Analysis 1) □ □□□□LLaMA □□ LLaMA □□□ □□□ □□□cross-domain consistent features가 □□□□가?

What & How

LLaMA SAE (32K features/layer): IC vs SM □paradigm □□decision-point BK Cohen's d □ □□ □ paradigm □□□□가ㅌ □ □□□active features(sign-consistent) □geometric mean |d| ≥ 0.2 □strong features □ □□.

Result → RQ1, RQ2

Layer	Active	Sign-consistent	%	Strong (d≥0.2)
L0	140	75	53.6%	52
L4	634	362	57.1%	261
L8	656	353	53.8%	240
L12	660	342	51.8%	251
L16	705	370	52.5%	253
L20	645	343	53.2%	230
L26	791	412	52.1%	258
L30	881	436	49.5%	281
Total (all 32L)	—	**5,374**	~51-57%	**3,633**

Comparison: Gemma (3-paradigm) vs LLaMA (2-paradigm):

	Gemma IC+SM+MW	LLaMA IC+SM
Sign-consistency rate	33-45% per layer	50-57% per layer
Total strong features	665 (7 layers)	3,633 (32 layers)

Critical Assessment: Binomial Test Against Chance

2-paradigm sign-consistency □chance level □2×0.5² = **50%** (random signs □□ □□□ □□). □□□51-57% 가 chance □□ □□□□가?

Layer	Observed %	Binomial p (vs 50%)	Significant?
L0	53.6%	0.224	NS
L4	**57.1%**	**0.0002**	*
L6	**57.6%**	**0.0001**	*
L8	53.8%	0.028	*
L10	50.0%	0.516	NS
L12	51.8%	0.185	NS
L20	53.2%	0.058	NS
L22	50.6%	0.397	NS
L26	52.1%	0.128	NS
L30	49.5%	0.632	NS

16 □layer □ □□□□**layer □3 □**(L4, L6, L8) □ □□□□, **13 □layer □□chance level** □ □□ □가.

Interpretation

LLaMA IC-SM raw sign-consistency □ □□□chance level □ □□□□. Gemma 3-paradigm (L18+ □□p < 5e-3, chance=25%) □ □□□□□ □□□ □ □□□ □□□

1. 2-paradigm vs 3-paradigm chance gap: 2-paradigm chance = 50%, 3-paradigm chance = 25%. 가ㅌ □absolute % □□3-paradigm □□ □□ □ □□□□□

2. "Strong" filter □ □□□ Raw sign-consistency □chance □□□□□, **geo_mean |d| ≥ 0.2** □**3,633** □ **"strong" features** □**effect size**가 □□□□□□ Random labels □□|d| 가ㅅ □ □□□□□, strong features □sign-consistency □raw % □□ □ □□ □□ □□□formal permutation test □□ □□ □□□□□ □□□

3. Cross-domain transfer (Analysis 7) □ □**strong** □ □□ Sign-consistency □feature-by-feature □ □□□□noise □ □□□□□, classifier transfer (SM→IC L30: AUC=0.685, p=0.000) □ □□feature space □joint distribution □ □□□□□ □robust □cross-domain □□□□

□□ LLaMA IC-SM □□cross-domain signal □ □□□□□, sign-consistency □□□ □□ □□□□, transfer AUC (Analysis 7)가 □ □□□ □ □□ □□□□

Analysis 9: Gemma SM Prompt Component Analysis (G/M/W/P/R)

Why

RQ3 □Fixed/Variable □□□ □□□□ □□□ □□□ □□□ Gemma SM V4role □5가 □prompt component □ □□ G(Goal), M(Mood), W(Warning), P(Persona), R(Reset). 가ㄱ component가 BK rate □neural activation □ □□□ □□ □□가?

What & How

SM 3,200 games □가ㄱ component □ □□□ □□ Per-component:

- BK rate with vs without → behavioral effect
- comp_sig_neurons: □□component □ □□□□□ □□□□neurons
- comp_x_bk_interaction_neurons: component × outcome interaction □ □□□□neurons (component가 BK □□□ □□)
- amplifies_bk: comp_x_bk interaction □ □□□+ component가 BK □□□ □□□□neurons

Result → RQ3

BK Rate Effect:

Component	BK with	BK without	Ratio	Primary mechanism
G (Goal)	5.2%	0.2%	**20.75x**	Independent BK driver
M (Mood)	3.8%	1.6%	2.35x	Mild additive
W (Warning)	3.8%	1.6%	2.35x	Mild additive
R (Reset)	3.4%	2.0%	1.72x	Moderate, layer-dependent
P (Persona)	2.9%	2.5%	1.17x	Negligible

Neural Analysis (comp_x_bk_interaction_neurons across layers):

Component	L10	L18	L22	L26	L30	L33	Pattern
G	109	168	168	308	**552**	**592**	□□□□ □가
M	41	23	25	10	8	15	□□ □□
R	205	314	235	102	95	97	□가ㄴ → 가ㅁ □
W	183	70	23	1	0	0	□□□□ □□
P	53	39	38	8	5	3	□□ □□

amplifies_bk (BK-amplifying interaction neurons):

Component	L10	L22	L30	Interpretation
G	0	0	1	G □BK □ □□□□□ □□; BK □□□ □□□□ □□
R	18	63	15	R □BK □□□ □가ㄴ layer □□ □□
W	42	3	0	W □shallow layer □□ □□BK □□, deep □□ □□
M, P	2-12	3-6	0	□□□ □□

Interpretation → RQ3

G(Goal) component □ □□□ □□ BK rate 20.75 □ □가. □□□G □BK neural representation □ □□□□ □□□ (`amplifies_bk≈0`) — G □BK □ □□□□□ □□□□, □□BK □□□□ □□ □□□□ □□□ "Goal □ □□□ □□ □□ □□가 □□□□"

R(Reset) component □ □가ㄴ **layer BK** □□ L22 □□63 □BK-amplifying neurons — R □ □□ □□ □□□□ □□ BK □□□ □□□□ □□ L22 □□ □가ㅁ (deep layers □□□G □ □□ □□□ □□).

W(Warning) □shallow-specific □□ Warning □ □□ □□ □□(L10: 42 amplifying neurons) □□□BK □ □□□□□□, deep layers □□□ □□□□ Warning □behavioral level □□□ □□□□deep semantic level □□□BK □ □□□ □□ □□□□ □

RQ3 □□(Gemma): □□□□ □□G가 BK □ □□□□□ □□□□□, G □ □□BK □□□□□ □□□□ □□ □□□□□BK □ □□□□

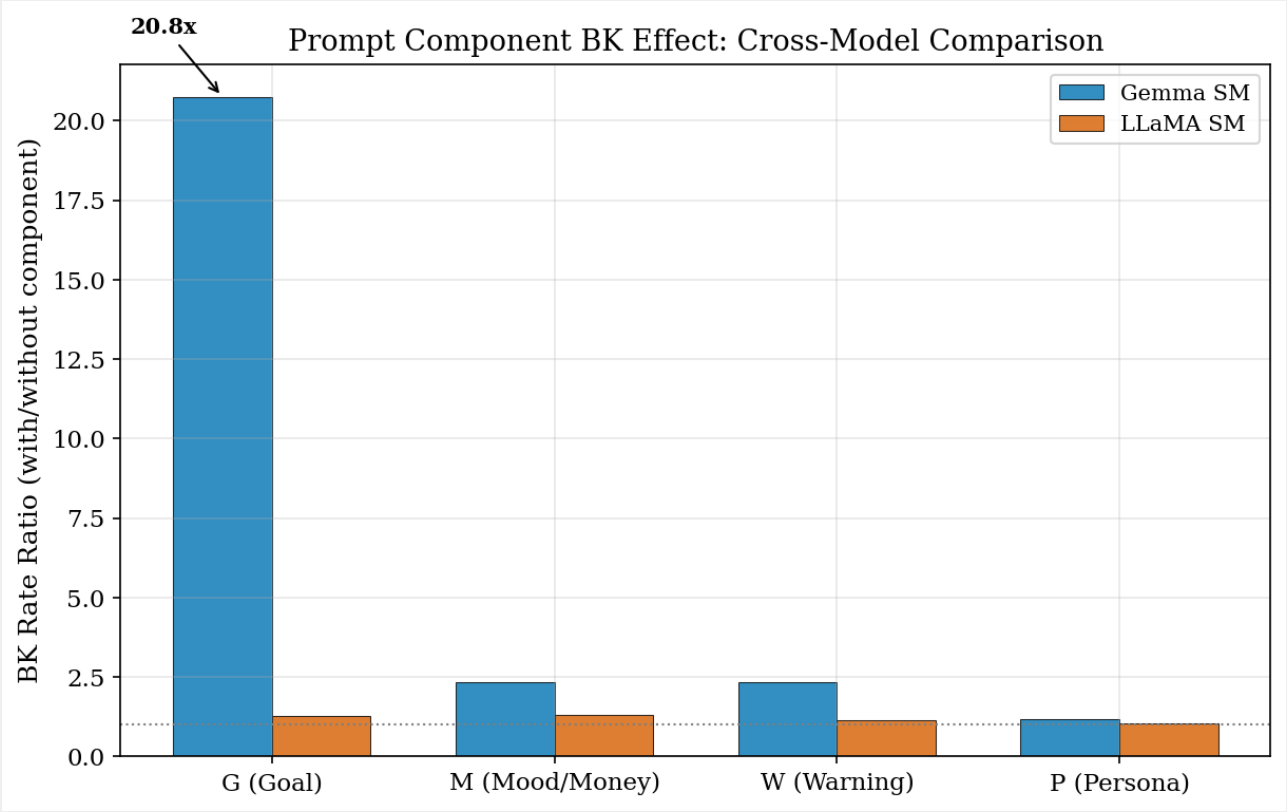

Figure 5: Prompt component BK effect — dramatic cross-model difference for G (Goal)

Figure 5 □□ G(Goal) component가 Gemma □□20.8x BK □가 (blue, dominant) □□□LLaMA □□□1.26x (orange, mild). 가ㅌ □prompt가 □□□ □□16 □ □□ □□□ □□ M, W, P □ □ □□ □□□□ □□

Analysis 9b: LLaMA SM Prompt Component Analysis (SAE Feature Level)

Why

Analysis 9 □□Gemma SM □prompt component □ □□□□ LLaMA SM □□prompt_conditions □ □□(G/M/H/W/P combinatorial). □□□cross-model □□가 □□

Note: LLaMA SM hidden states가 v9.4 □□5 layers (L8,12,22,25,30) □ □□ □□□□□□□, Analysis 9b □SAE feature level □ □□(32 layers □□ □□□□). Gemma SM □hidden state neuron (3,584-dim, 42 layers) □□ Absolute counts □□ □□ □가; □□(%) □ □□

What & How

```
LLaMA SM 3,200 games. Prompt conditions: combinatorial (BASE, G, GM, GHP, GMHWP □ 32□).
□ component (G/M/H/W/P) □□□ 1,600 vs 1,600 □□.
Per-component, per-layer [L8, L12, L20, L25, L30]:
  - BK rate ratio (with/without)
  - Per-SAE-feature: feature ~ outcome + component + outcome×component
  - interaction_sig: p(interaction) < 0.01
  - amplifies_bk: interaction sig AND component increases |BK-feature effect|
```

Result → RQ1, RQ3

BK Rate Effect (cross-model comparison):

Component	Gemma SM ratio	LLaMA SM ratio	Cross-model
G (Goal)	**20.75x**	**1.26x**	Gemma >>>>> LLaMA
M (Mood/Money)	2.35x	**1.31x**	Gemma > LLaMA
W (Warning)	2.35x	1.15x	Gemma > LLaMA
H (Hint)	—	1.05x	LLaMA only
R (Reset)	1.72x	—	Gemma only
P (Persona/Positive)	1.17x	1.06x	□ □negligible

G component □ □□□cross-model □□□ Gemma □□G □BK □20.75 □ □가 □□□□, LLaMA □□□1.26 □□ □□ 가ㅌ □ Goal prompt가 □□□ □□ □□□ □□ □□□ □□□ □□

Neural Analysis: interaction + amplifies_bk (% of active SAE features):

Component	L8 int%	L12 int%	L30 int%	L30 amp%	Pattern
G	45.5%	61.5%	**63.0%**	8.2%	□□□□interaction □가, but amp □□
M	**55.4%**	48.2%	**51.2%**	**14.5%**	□layer □□highest amplification
H	25.9%	31.2%	33.3%	7.4%	□□ □□
W	36.2%	37.2%	19.2%	5.8%	shallow □□가ㅈ, deep □□ □
P	26.0%	27.1%	31.2%	4.8%	□□ □□

Interpretation → RQ1, RQ3

Cross-model prompt component □□□□ □□□ □□ □□:

1. G □ □□□ □□가 □□ □□□ Gemma SM □□G □BK 20.75x (□□□ □□) □□□, LLaMA SM □□G □1.26x (mild). □□ RLHF/RLAIF training □ □□□, □ □□□Goal prompt □ □□ □□ □□ □□ □□□ □□□ □

2. Neural mechanism □G □□ □□, M □□ □□ □ □□ □□G □interaction neurons가 layer □ □□ □가 □□ amplifies_bk가 □□(Gemma: amp≈0, LLaMA: amp=8.2%). □**G** □ □ □□□□**BK** □ □□□□□ □□□□ □□가 □□□ □□, LLaMA □□M(Money framing) □highest BK amplifier (14.5% at L30) — Gemma □□M □ □□ □□LLaMA가 financial framing □ □ □가ㅁ □□ □□□□ □□

3. W(Warning) □layer pattern □□□ □□□□□W □shallow layers □□ □□가 가ㅈ □□deep □□ □□□(Gemma: L10=183→L30=0; LLaMA: L8=36%→L30=19%). Warning □ □□ □□□□□ □□□□□ □□□cross-model □ □□

Analysis 8b: LLaMA Within-Bet-Type Cross-Domain Consistency

Why

Gemma □□bet-type □cross-domain consistency □ □□□□(Analysis 4 Supplementary: Fixed IC+MW 57.2%, Variable IC+SM 61.6%). LLaMA □ □□□ □□

Result → RQ2

Bet-type	Paradigms	Active features	Sign-consistent	%	d_correlation	p
Fixed	IC + SM	630	351	55.7%	**0.209**	1.2e-7
Variable	IC + SM	719	309	43.0%	**-0.097**	0.009

Critical issue — Variable □□negative d_correlation:

LLaMA Variable games □IC-SM Cohen's d □□□ □□(r=-0.097). IC □□BK-promoting □features가 SM □□ BK-inhibiting □ □□ □□Variable betting □□ □paradigm □BK representation □ □□ □□□□ □□□□ □□

BK rate artifact □□: SM Variable BK rate □IC □ □□□□(9.6%) subsampling □ □50-bootstrap d_correlation □ □□□→ **mean=-0.075, 95% CI=[-0.179, +0.032], 18% □positive**. BK rate matching □□□negative d_correlation □ □□□□□, □□**BK rate artifact**가 □□ □□□**representational** □□ IC □SM □□Variable betting □BK □□□ □□□□□ □□□

□□, **Fixed □□positive d_correlation (0.209, p=1.2e-7)** □Fixed betting □□IC-SM cross-domain consistency가 □□□□ □□ Fixed SM BK=0.4% (n=7) □ □□ □□□, □□□□IC Fixed □ □□

Synthesis: □가 □ □□ □□□ □□ □

RQ1: Gemma □LLaMA □□ □□□ □□□ □□가?

Yes — □□□ □□□□ □□□□, □□□ □□□ □□□□.

□□	가ㅈ□	□□
BK □□ □□	가ㅈ	Gemma 0.976 vs LLaMA 0.974 (SM); Gemma 0.960 vs LLaMA 0.954 (IC)
BK-inhibiting □□(L30)	가ㅈ	Gemma L30: p=0.000; LLaMA SM L30: p=0.001; LLaMA IC L30: p=0.000
BK-inhibiting □□(L22)	□	LLaMA SM L22: BK+ > BK- (p=0.897 NS) — mid-layer □□□ □□ □ □
Factor decomposition	가ㅈ	Outcome-significant: Gemma 65.2%, LLaMA 69.5% — □□□ □□□BK □
IC↔SM transfer □□□	□	□ □□ □□IC→SM layer-dependent, SM→IC □ □□
LLaMA sign-consistency	□	2-paradigm chance=50%; 16 layers □3 □□ □□□
IC BK rate □□	□□	Gemma Fixed>>Variable, LLaMA Variable>Fixed — 가ㅌ □paradigm □□ □□ □□
SM BK rate □□	□□	Gemma SM 2.7% vs LLaMA SM 36.4% — 27x □□, 가ㅌ □ □□□□ □□ □□ □□

RQ2: □□□□ □□□□ □□□ □□ □□□ □□가?

Yes — 가ㅈ □ □□(Gemma), □□□ □□(LLaMA).

□□	가ㅈ□	□□
Gemma IC→MW transfer	가ㅈ	L18: 0.932, L22: 0.876, L26: 0.883 (all p=0.000)
Gemma SM→MW transfer	가ㅈ	0.789-0.867 (all p=0.000)
Gemma IC→SM transfer	□	Layer-dependent: L26=0.913 but L22=0.499(NS)
Gemma 3-paradigm SAE consistency	가ㅈ	L18+ 33-45% (p < 5e-3 vs 25% chance); 665 strong features
Factor decomposition	가ㅈ	65.2% SAE features outcome-significant after controlling bet_type+paradigm
LLaMA Factor decomposition	가ㅈ	69.5% SAE features outcome-significant (Gemma 65.2% □ □□)
LLaMA IC→SM transfer (L25)	□	0.783 (p=0.000) □□layer □□□
LLaMA SM→IC transfer	□	L30=0.685, L12=0.662 (all p=0.000)
LLaMA Fixed cross-domain	□	55.7% sign-consistent, d_corr=0.209 (p=1.2e-7)
LLaMA Variable cross-domain	□□	d_corr=-0.075 after BK rate matching (18% positive) — BK □□□ genuinely □□

RQ3: □□□□ □□□ □□ □□□□가?

Yes — BK representation □□□ □□ □□□□, □□□BK □□ □□□ □□□□. □, □□□ □□ □□□ □□□□ □□ □□□.

□□	가ㅈ□	□□
IC Fixed/Variable cos □□ (Gemma)	□	Deep layers cos > 0.33-0.44; □Variable BK=14 □
IC Fixed/Variable cos (LLaMA)	가ㅈ	□**layer cos > 0.81** (both groups n≥65); Gemma shallow negative □ sample size artifact □□
Shared BK neurons (□ □□IC)	가ㅈ	Gemma 1,182 (33%) vs **LLaMA 1,501 (36.6%)** — cross-model □□
Balance confound control (□ □□)	가ㅈ	Gemma 99.7%, LLaMA 101% — □ □□ □□balance confound □□
G □□BK □□(□ □□)	가ㅈ	□ □□ □□G □amplifies_bk □□(Gemma≈0, LLaMA=8.2%) — G □BK □□ □□ □ □ □□ □□
G behavioral effect □□ □□	□□	Gemma G=20.75x vs LLaMA G=1.26x — 가ㅌ □**prompt**가 □□□□**16** □ □ □ □□
W shallow pattern (□ □□)	□	□ □□ □□W □shallow □□가ㅈ, deep □□ □□— cross-model □□
M amplification □□	□	LLaMA M=14.5% BK amplification (highest), Gemma M≈0 — financial framing □가ㅁ □ □□

□□: " □□ □□□ □□ □□" □ □□

LLM □□□□□" □□ □□(bankruptcy representation)" □ □□□□□, □ □□□ □□□ □□□□□□ □□□□.

□□□ □□(permutation/binomial test □ □□):

1. BK □□□ □□□ □□: 67-71% SAE features가 bet_type·paradigm □□ □□□BK-significant (perm_p=0.000, null ~1%)

2. Deepest layer (L30) BK-inhibiting □□: □ □□, □paradigm □□□□BK- > BK+ (binomial p<0.001). □, **mid-layer (L22) □□□LLaMA SM □□ □□□□□□ □□**(p=0.897) — " □□ □□" □□□L30+ □□ □□

3. Cross-domain transfer: Gemma IC→MW 0.93, SM→MW 0.87 (all p=0.000). LLaMA SM→IC 0.685 (p=0.000). □, **IC→SM □ □ □□ □□layer-dependent**

4. High classification AUC: Gemma 0.960-0.976, LLaMA 0.954-0.974 (perm_p=0.000)

□□□□ □□ □□(□가 □□ □□ □□ □□ □□):

1. "BK-inhibiting dominance" at all layers: L22 □□LLaMA SM □BK+가 □ □□→ layer-specific claim □□ □□□□

2. Variable betting cross-domain consistency: LLaMA IC-SM Variable d_corr=-0.075 (BK rate matching □□□negative, 18% □positive) → **Variable □□IC □SM □BK □□□genuinely □□□**

3. Cross-model effect size comparability: Gemma mean|d|=0.571 vs LLaMA 0.209 — **sample size confound** (87 vs 1164) □□ □□ □□ □가

□□: BK representation □" □□" (67-71% independent, high AUC) □"deepest-layer □□" (inhibiting □□) □ □□□□ □□□□ □□□□. □□□

- □□ □□□**BK □trigger □□**가 □ □□□□ □□□□ □□□(G prompt: 20.75x vs 1.26x; IC Fixed/Variable: □□□)

- **Cross-domain transferability** □ □□□□□(IC→MW 가ㅈ, IC→SM □□□, Variable □cross-domain □□)
- "Neural hardware" □ □□□L30+ deepest layers □ □□□□, □가ㄴ layers □□□ □□ □□□ □□□ □□□ □□□□

Critical Assessment: □□□가ㅈ □□ □□

Permutation-Validated Claims

Claim	Test	Result
Factor decomposition 67-71%	100-perm shuffle	p=0.000 (null 1%) ✅
BK-inhibiting dominance L30	Binomial test	Gemma p=0.000, LLaMA SM p=0.001, LLaMA IC p=0.000 ✅
BK-inhibiting dominance L22	Binomial test	LLaMA SM p=0.897 ❌ — mid-layer □□ □□□
Variable cross-domain d_corr	BK rate matching + bootstrap	mean=-0.075, CI[-0.179,+0.032] ❌ — genuine □□
Classification AUC	100-perm shuffle	p=0.000 ✅
Gemma 3-paradigm consistency (L18+)	Binomial vs 25%	p < 5e-3 ✅

Cross-Model Effect Size Confound

	Gemma SM	LLaMA SM		
n_BK	87	1,164		
mean	d		0.571	0.209
% d≥0.3	68.8%	23.2%		

Gemma □ □□|d| □**small sample inflation** □□□가 □□□ □□(N=87 → noisy estimates → inflated absolute values). LLaMA □ □□|d| □larger N □□true effect □ □□□ □□ □ □□ □□□**cross-model effect size** □ □□ □□ □가. Classification AUC (sample size □ □ □가ㅁ)가 □reliable □ □□ □□.

□□□ □□□ □□

□□	□□	□□
Analysis 2 (Variable BK direction)	n_var_bk=14 (Gemma IC)	Cosine similarity □single estimate, CI □□, t-test power □□ □□
Analysis 8 (LLaMA sign-consistency)	2-paradigm chance=50%, 13/16 layers NS	Raw sign-consistency □cross-domain □□□ weak evidence
Analysis 1 (L10, L12)	3-paradigm binomial test NS (p > 0.05)	Shallow layer cross-domain signal □ noise □ □ □□

BK Rate □□□

□□	**BK rate □□**	□□
Gemma IC Fixed vs Variable	19.75% vs 1.75% (11x)	Fixed가 □□ □□— Variable □" □□ □□" □□□□IC □□□ □□
Gemma SM vs LLaMA SM	2.7% vs 36.4% (13x)	Cross-model AUC □□가 □□□□□ □□; base rate가 □□□ classifier □□□ □□
IC→SM transfer	IC 8.9% BK vs SM 36.4% (LLaMA)	SM→IC transfer □ □□AUC가 BK prevalence □□□ □□□□□ □ □□

IC Fixed/Variable BK Rate □□

Gemma IC □□Fixed betting □Variable □□11 □ □□BK rate □ □□□. □□SM □"Variable → □ □□BK" □□□ □□□. IC □□ Fixed bet □" □ □□□ □□ □□ □□" □, Variable □" □□ □□가 □" □ □□□□. Fixed가 □ □□□ □□: □□ □□□□□ □□ □ □□ □□□ □가 □→ □□ □□ □□ Variable □□□loss-sensitive □□□□ □□ □□가 □ □ □□□□ □□□**BK** □□□**cross-paradigm** □□ □ □□□ □□□.

□□ □□ □□

□□ □□□□FDR correction □□Bonferroni correction □ □□□□□ □□□□ □□□ □□:

- Analysis 3: 3,584 neurons × individual t-tests → uncorrected $p < 0.01$ □□→ false positives ~36 □ □□
- Analysis 4: 581 SAE features × regression → uncorrected $p < 0.01$ □□→ false positives ~6 □ □□
- Analysis 9: 3,584 neurons × 5 components × 6 layers → □□ □□tests

Analysis 1, 5, 7 □permutation test □ □□□□ □ □□□ □□□□□ □□ □□□ □□□ □□□effect size □□(Cohen's d) □ primary □, p-value □secondary □ □□□□ □□.

Limitations

□□□ □□

1. ~~LLaMA hidden states □□□~~→ □□□(v9.4): Decision-point hidden states □5 layers [L8,L12,L22,L25,L30] □□ □□□ Analysis 2, 3 □LLaMA □□ □□ □□ □, □□32 layers가 □□5 layers □ □□.

2. LLaMA MW □□□: 3-paradigm cross-model □□ □가.

3. Gemma IC Variable BK=14 □ Bet-type □□ □□ □□□power □□ □□.

4. MW Variable BK=4 □ b2a □□□(n=5) □□, MW bet-type □□ □가.

□□ □□□ □□

5. LLaMA sign-consistency □□(Analysis 8): 2-paradigm chance=50% □□13/16 layers NS. Sign-consistency □2-paradigm □□ □□ □□, transfer AUC가 □ □□□ □ □□ □□.

6. LLaMA Variable cross-domain d_corr □□(Analysis 8b): IC Variable BK=9.6% vs SM Variable BK=72.3% □ □□□base rate □□가 cross-domain comparison □ □□.

7. IC→SM Transfer □□□: □ □□ □□layer-dependent (Gemma L22 □□/L26 □□; LLaMA L25 □□□).

8. □□□ □□□: □□ □□□correlational. Activation patching □□causal validation □□.

9. □□□□ □□□: Analysis 3, 4, 6b, 9, 9b □□FDR correction □□□.

□□ □□

10. BK rate heterogeneity: Gemma IC Fixed 19.75% vs Variable 1.75%; LLaMA IC Fixed 8.1% vs Variable 9.6% (□□); Gemma SM 2.7% vs LLaMA SM 36.4% (13x). Cross-model/cross-paradigm □□ □base rate confound □□.

11. Prompt component effect □ □□ □□□: G component가 Gemma □□20.75x, LLaMA □□1.26x. "Prompt condition invariance" □neural structure □□ □□□□□, behavioral effect □□ □□□□ □□.

12. SAE feature vs hidden state neuron □□: Gemma Analysis 9 □hidden state neurons (3,584-dim) □□, LLaMA Analysis 9b □SAE features (32K, active ~800-1400) □□. Absolute counts □□ □□ □가; % □□ □□.

Next Steps

1. ~~LLaMA Hidden State~~ □□→ □□**(v9.4)**. Analysis 2, 3 □□ □□ □□.

2. Activation patching (□□ □□**)**: Gemma 665 cross-domain strong features □top candidates □ □□ causal validation.

3. LLaMA MW □□: 3-paradigm cross-model comparison □□.

4. IC→SM Transfer □□□□: L26 □□/ L22 □□□ □□ □□.

5. FDR correction □□: □□per-feature/per-neuron tests □FDR correction □□.

6. G component cross-model □□ □□: G가 Gemma □□20.75x, LLaMA □□1.26x □ □□— □ □□□G prompt □□ □□ □□(□□layer □□diverge □□가).

7. BK rate confound □□: IC-SM transfer □□balanced subsampling □□SMOTE □base rate □□ □ □□□.